

ULTIMATE EXCEL FORMULAS & TOOLS MASTER DIRECTORY

Complete Comprehensive Reference Sheet — Personalized for PRAVEEN VIJAYAPURAM

1. Essential Excel Data Tools

Pivot Table

Definition: An advanced built-in summarization tool that automatically groups, sorts, counts, totals, or averages data stored in one large table.

Core Usage: Instantly transforming millions of rows of transaction histories into structured summaries (e.g., viewing total quarterly sales grouped by individual regions and product categories).

Key Layout

Quadrants: **Filters:** Slices the whole report; **Columns:** Side-by-side data fields; **Rows:** Vertical breakdown blocks; **Values:** Numbers to calculate.

Real-World Case: Dragging 'Region' into Rows, 'Year' into Columns, and 'Sales' into Values to instantly generate a complete multi-year sales comparison dashboard.

Data Validation

Definition: An input control restriction rule feature that governs exactly what text patterns, ranges, numbers, or dropdown lists can be typed into a cell.

Core Usage: Preventing human error and maintaining layout structural consistency across collaborative operational worksheets.

Real-World Case: Creating an immutable drop-down choice list constraint selection containing only { "Pending" , "Approved" , "Rejected" } so that teammates cannot input accidental typing variations.

Remove Duplicates

Definition: A data cleaning function that scans specific chosen table rows and drops matching identical repeat data combinations permanently.

Core Usage: Cleaning raw system exports to extract crisp unique lists of item arrays, customer logs, or system indicators.

Real-World Case: Selecting an active raw client acquisition data sheet containing multiple re-entries and purging matching duplicates using 'Email ID' as the target unique master key.

2. Primary Mathematical Formulas

SUM

Definition: Adds all the numerical values in a selected range of cells.

Usage: Finding grand totals of financial numbers, quantities, or invoice lines.

Structure: `=SUM( range )`

Example: `=SUM( C2 : C20 )`

PRODUCT

Definition: Multiplies all the numbers given as arguments.

Usage: Multiplying dimensions or compounding rates without using manual asterisk symbols.

Structure: `=PRODUCT(number1, [number2], ...)`

Example: `=PRODUCT(A2, B2)`

ABS

Definition: Returns the absolute value of a number (converts negative numbers to positive numbers).

Usage: Finding the variance or distance between two metrics regardless of which one is larger.

Structure: `=ABS(number)`

Example: `=ABS(-150)`

3. Rounding Formulas

ROUND

Definition: Rounds a number to a specified number of decimal digits based on standard mathematical rules.

Usage: Cleaning up raw, messy financial decimals into standard currency formats.

Structure: `=ROUND(number, num_digits)`

Example: `=ROUND(15.678, 2)`

ROUNDUP

Definition: Forces a number upwards, away from zero, regardless of the decimal value.

Usage: Calculating the minimum number of whole boxes, shifts, or assets needed.

Structure: `=ROUNDUP(number, num_digits)`

Example: `=ROUNDUP(12.1, 0)`

ROUNDDOWN

Definition: Forces a number downwards, toward zero, dropping any remaining decimal fraction.

Usage: Tracking strictly completed milestones, completed hours, or fully packed units.

Structure: `=ROUNDDOWN(number, num_digits)`

Example: `=ROUNDDOWN(45.99, 0)`

INT

Definition: Rounds a number down to the nearest whole integer.

Usage: Stripping time stamps away from dates or removing decimal configurations completely.

Structure: `=INT(number)`

Example: `=INT(45.95)`

4. Statistical & Counting Formulas

AVERAGE

Definition: Calculates the arithmetic mean of a selected range of numbers.

Usage: Tracking core performance baselines like average daily sales or average login times.

Structure: `=AVERAGE( range )`

Example: `=AVERAGE( D2 : D50 )`

COUNT

Definition: Counts the number of cells in a range that contain numerical values only.

Usage: Auditing datasets to ensure transactional rows have quantities or financial items filled in.

Structure: `=COUNT( range )`

Example: `=COUNT( C2 : C100 )`

COUNTA

Definition: Counts the number of cells in a range that are not empty (text, numbers, spaces, errors).

Usage: Identifying the total active row volume in a data table.

Structure: `=COUNTA( range )`

Example: `=COUNTA( A2 : A100 )`

COUNTBLANK

Definition: Counts the number of completely empty cells in a specified range.

Usage: Scanning an operations tracker to pinpoint missing fields or incomplete entries.

Structure: `=COUNTBLANK( range )`

Example: `=COUNTBLANK( B2 : B100 )`

MAX

Definition: Returns the highest (maximum) number from a given set of values.

Usage: Identifying the highest individual sale, peak volume hour, or maximum target achieved.

Structure: `=MAX( range )`

Example: `=MAX( E2 : E50 )`

MIN

Definition: Returns the lowest (minimum) number from a given set of values.

Usage: Tracking lowest operational cost, minimum production run, or earliest check-in time.

Structure: `=MIN( range )`

Example: `=MIN( E2 : E50 )`

LARGE

Definition: Returns the k-th largest value in a dataset (e.g., the 2nd highest).

Usage: Isolating relative rankings, like finding the runner-up sales figures without sorting.

Structure: `=LARGE(array, k)`

Example: `=LARGE(A2:A50, 2)`

SMALL

Definition: Returns the k-th smallest value in a dataset (e.g., the 3rd lowest).

Usage: Tracking low-end markers, like identifying the third fastest turnaround time.

Structure: `=SMALL(array, k)`

Example: `=SMALL(A2:A50, 3)`

5. Text Manipulation & Data Cleaning

TRIM

Definition: Removes all leading, trailing, and duplicate spaces from text strings.

Usage: Crucial first step for cleaning text inputs so structural formulas do not break.

Structure: `=TRIM(text)`

Example: `=TRIM(B2)`

PROPER

Definition: Capitalizes the first letter of each word and forces remaining letters to lowercase.

Usage: Standardizing employee names, vendor records, or city titles for neat reporting.

Structure: `=PROPER(text)`

Example: `=PROPER("bengaluru, INDIA")`

UPPER

Definition: Converts all letters in a text string to uppercase.

Usage: Creating standard codes like PAN numbers, IFSC codes, or status acronyms.

Structure: `=UPPER(text)`

Example: `=UPPER("ka68")`

LOWER

Definition: Converts all letters in a text string to lowercase.

Usage: Formatting system usernames, database entries, or email lists consistently.

Structure: `=LOWER(text)`

Example: `=LOWER("PRITI@brightpath.com")`

CONCAT

Definition: Links text fields together into a single cell.

Usage: Generating composite values, like stringing together addresses or full names.

Structure: `=CONCAT(text1, text2, ...)`

Example: `=CONCAT(A2, " ", B2)`

TEXTJOIN

Definition: Links text fields while inserting a specified delimiter between each value.

Usage: Creating clean, readable summaries of multiple items in a single cell.

Structure: `=TEXTJOIN(delimiter, ignore_empty, text1, ...)`

Example: `=TEXTJOIN(", ", TRUE, A2, B2, C2)`

LEFT

Definition: Extracts a specified number of characters starting from the far-left side of a text string.

Usage: Extracting prefix codes, country calling numbers, or primary category labels.

Structure: `=LEFT(text, [num_chars])`

Example: `=LEFT(A2, 5)`

RIGHT

Definition: Extracts a specified number of characters starting from the far-right side of a text string.

Usage: Pulling out file extensions (like .xlsx), serial endings, or account suffixes.

Structure: `=RIGHT(text, [num_chars])`

Example: `=RIGHT(B2, 4)`

MID

Definition: Extracts a specific number of characters from the middle based on a starting position.

Usage: Parsing structured values to isolate area codes or internal serial fragments.

Structure: `=MID(text, start_num, num_chars)`

Example: `=MID("BRIGHTPATH", 3, 4)`

LEN

Definition: Counts the total number of characters (including spaces) inside a target cell.

Usage: Validating input lengths to make sure phone numbers or IDs have correct digit counts.

Structure: `=LEN(text)`

Example: `=LEN(C2)`

SUBSTITUTE

Definition:	Replaces specific matching text string patterns with new text.
Usage:	Updating system codes or fixing recurrent formatting typos across spreadsheets.
Structure:	<code>=SUBSTITUTE(text, old_text, new_text, [instance_num])</code>
Example:	<code>=SUBSTITUTE(A2, "-", "/")</code>

6. Logical Engine Formulas

IF

Definition:	Evaluates an expression and outputs one specific value if TRUE, and another if FALSE.
Usage:	Categorizing transactional rows into discrete business paths based on rules.
Structure:	<code>=IF(logical_test, value_if_true, value_if_false)</code>
Example:	<code>=IF(A2>5000, "High Tier", "Standard")</code>

IFS

Definition:	Evaluates multiple conditions sequentially and returns the value of the first TRUE condition.
Usage:	Managing complex, tiered logic loops (like grade scales) without nested IF blocks.
Structure:	<code>=IFS(condition1, value1, condition2, value2, ...)</code>
Example:	<code>=IFS(A2>=90, "A", A2>=80, "B", A2>=70, "C")</code>

AND

Definition:	Tests multiple conditions and returns TRUE only if every single condition inside is met.
Usage:	Used when a candidate or data row must clear multiple strict rules simultaneously.
Structure:	<code>=AND(logical1, logical2, ...)</code>
Example:	<code>=AND(A2="Graduate", B2>=21)</code>

OR

Definition:	Tests multiple conditions and returns TRUE if at least one of the conditions is met.
Usage:	Used when there are multiple alternative ways to qualify for a rule.
Structure:	<code>=OR(logical1, logical2, ...)</code>
Example:	<code>=OR(C2="Bengaluru", C2="WFH")</code>

NOT

Definition:	Reverses the logical value of its argument. Turns TRUE into FALSE, and vice versa.
Usage:	Used to filter out specific items or create an "exclude" rule in tracking.
Structure:	<code>=NOT(logical)</code>
Example:	<code>=NOT(E2="Blacklisted")</code>

IFERROR

Definition:	Intercepts system error codes in formulas and outputs a clean fallback value instead.
Usage:	Keeping dashboards clean by masking ugly errors like #N/A or #DIV/0!.
Structure:	<code>=IFERROR(value, value_if_error)</code>
Example:	<code>=IFERROR(VLOOKUP(A2, B:C, 2, FALSE), "Not Found")</code>

7. Conditional Aggregations & Summaries

COUNTIF

Definition:	Counts the cells within a target range that match one single criteria.
Usage:	Tracking specific volumes, such as counting how many rows are marked as "Pending".
Structure:	<code>=COUNTIF(range, criteria)</code>
Example:	<code>=COUNTIF(A:A, "Pending")</code>

COUNTIFS

Definition:	Counts the cells within target ranges that match multiple criteria simultaneously.
Usage:	Tracking intersecting variables, such as counting rows from "Bengaluru" with "Urgent" status.
Structure:	<code>=COUNTIFS(criteria_range1, criteria1, criteria_range2, criteria2, ...)</code>
Example:	<code>=COUNTIFS(A:A, "Bengaluru", B:B, "Urgent")</code>

SUMIF

Definition:	Sums numerical cell values in a range matching a single criteria.
Usage:	Finding performance numbers for a single variable, like total sales in the "North" region.
Structure:	<code>=SUMIF(range, criteria, [sum_range])</code>
Example:	<code>=SUMIF(A:A, "North", C:C)</code>

SUMIFS

Definition:	Sums numerical cell values across a chosen range based on multiple distinct criteria blocks.
Usage:	Core reporting engine for rolling up specific financial figures based on multi-variable filters.
Structure:	<code>=SUMIFS(sum_range, criteria_range1, criteria1, criteria_range2, criteria2, ...)</code>
Example:	<code>=SUMIFS(C:C, A:A, "North", B:B, "Electronics")</code>

AVERAGEIF

Definition:	Calculates the mathematical average of cells that meet a single specific condition.
Usage:	Isolating baseline performance trends, like identifying average delivery times for a vendor.
Structure:	<code>=AVERAGEIF(range, criteria, [average_range])</code>
Example:	<code>=AVERAGEIF(E:E, "Completed", D:D)</code>

AVERAGEIFS

Definition: Calculates the mathematical average of cells that meet multiple conditions.

Usage: Fine-tuning performance audits across intersecting operational metrics.

Structure: `=AVERAGEIFS(avg_range, criteria_range1, criteria1, criteria_range2, criteria2, ...)`

Example: `=AVERAGEIFS(D:D, E:E, "Completed", F:F, "High Priority")`

8. Lookups, References & Data Matching

VLOOKUP

Definition: Searches vertically down the first column of a table to fetch matching info from the same row.

Usage: Pulling records from central master lists using a unique key like an Employee ID.

Structure: `=VLOOKUP(lookup_value, table_array, col_index, [range_lookup])`

Example: `=VLOOKUP(A2, Sheet2!A:D, 3, FALSE)`

HLOOKUP

Definition: Searches horizontally across the top row of a table to fetch matching info from the same column.

Usage: Pulling records from horizontally structured sheets, like tax tables or timeline columns.

Structure: `=HLOOKUP(lookup_value, table_array, row_index, [range_lookup])`

Example: `=HLOOKUP(A2, Sheet2!A1:M4, 2, FALSE)`

INDEX

Definition: Returns the value of a cell at the intersection of a specific row and column number.

Usage: Acts as the extraction tool that pulls data out of a grid once coordinates are known.

Structure: `=INDEX(array, row_num, [column_num])`

Example: `=INDEX(B2:B100, 5)`

MATCH

Definition: Searches for an item in a range and returns its relative row or column position number.

Usage: Acts as a tracker to find out exactly which row or line a unique ID or name sits on.

Structure: `=MATCH(lookup_value, lookup_range, [match_type])`

Example: `=MATCH("Priti", A2:A100, 0)`

XLOOKUP

Definition: Searches a designated range for an item and extracts data from a separate destination range.

Usage: The modern, flexible VLOOKUP replacement. Searches in any direction and defaults to exact match.

Structure: `=XLOOKUP(lookup_value, lookup_array, return_array, [if_not_found])`

Example: `=XLOOKUP(A2, B:B, C:C, "Missing")`

9. Date & Time Operations

TODAY

Definition: Automatically displays the current system date in a cell. Updates dynamically on refresh.

Usage: Setting up deadline tickers or calculating how many days old an active ticket or pending task is.

Structure: `=TODAY()`

Example: `=TODAY()`

NOW

Definition: Displays the current system date along with the exact live time. Recalculates on sheet changes.

Usage: Generating precise login/logout stamps or tracking data entry submission times.

Structure: `=NOW()`

Example: `=NOW()`

YEAR

Definition: Extracts the individual year integer out of a standard calendar date cell.

Usage: Grouping transactional rows into master buckets by calendar year.

Structure: `=YEAR(date)`

Example: `=YEAR(A2)`

MONTH

Definition: Extracts the individual month number (1 to 12) out of a standard calendar date cell.

Usage: Breaking down operational timelines into monthly review buckets.

Structure: `=MONTH(date)`

Example: `=MONTH(A2)`

DAY

Definition: Extracts the day integer (1 to 31) out of a standard calendar date cell.

Usage: Isolating day-of-the-month trends in a tracking sheet.

Structure: `=DAY(date)`

Example: `=DAY(A2)`

DATEDIF

Definition: Calculates the exact difference in days, months, or years between two calendar dates.

Usage: Tracking project ages, counting employee tenures, or calculating age metrics.

Structure: `=DATEDIF(start_date, end_date, "unit")`

Example: `=DATEDIF(A2, TODAY(), "Y")`

NETWORKDAYS

Definition:	Returns the exact count of net working days between two dates, skipping weekends/holidays.
Usage:	Tracking fulfillment timelines or operational windows based on strict business working days.
Structure:	=NETWORKDAYS(start_date, end_date, [holidays])
Example:	=NETWORKDAYS(A2, B2, Sheet3!G1:G10)

10. Dynamic Arrays (Modern Excel Engines)

UNIQUE

Definition:	Extracts and outputs an ordered list of distinct, non-repeating values from a specified range.
Usage:	Instantly cleaning out duplicate entries to generate a clean, dynamic summary master list.
Structure:	=UNIQUE(array)
Example:	=UNIQUE(B2:B500)

FILTER

Definition:	Filters an entire data array based on a logic rule, dynamically outputting every matching record.
Usage:	Creating automated, live-updating sub-reports without clicking manual filter dropdowns.
Structure:	=FILTER(array, include, [if_empty])
Example:	=FILTER(A2:D100, B2:B100="Bengaluru", "No Records")

SORT

Definition:	Sorts the contents of a range or array dynamically based on a specific column index.
Usage:	Building live dashboards that keep records neatly arranged by values or dates automatically.
Structure:	=SORT(array, [sort_index], [sort_order])
Example:	=SORT(A2:C100, 3, -1)